LESSON 05A

Work changes energy

Systems • conservative forces • the surprising work done by friction

$$\Delta K + \Delta U = W_{ext}$$

The question is not just “What forces act?”

It is also: “What is inside our system?”

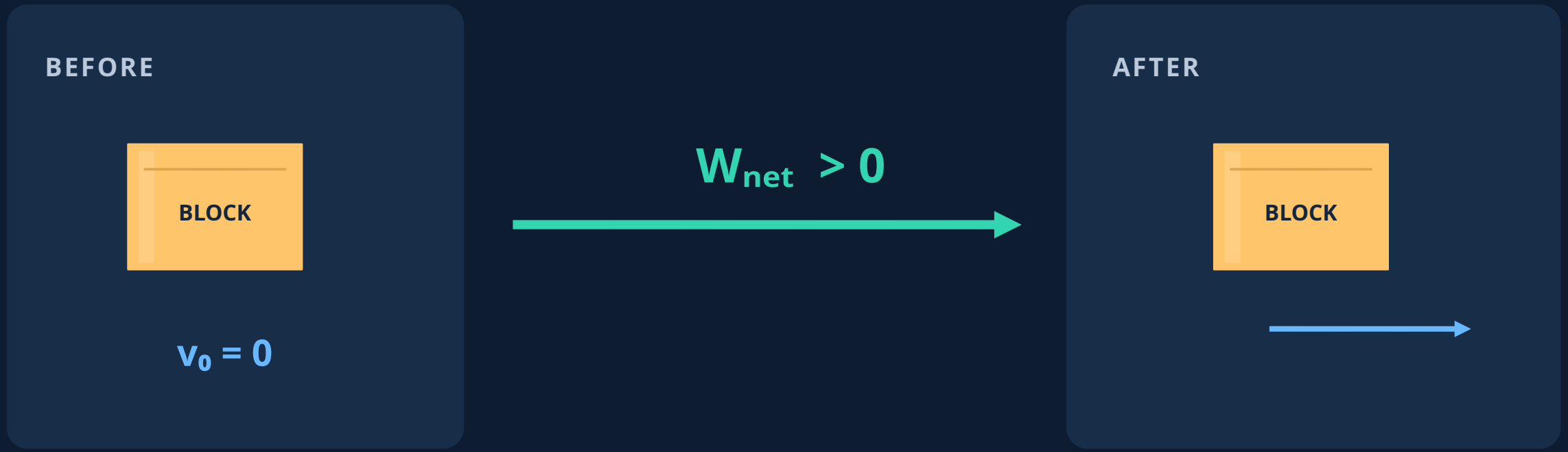


Physical work. Energy choices.

CORE IDEA

Work changes kinetic energy

For an object, add the work done by all the forces acting on it.



$$\Delta K = W_{net}$$

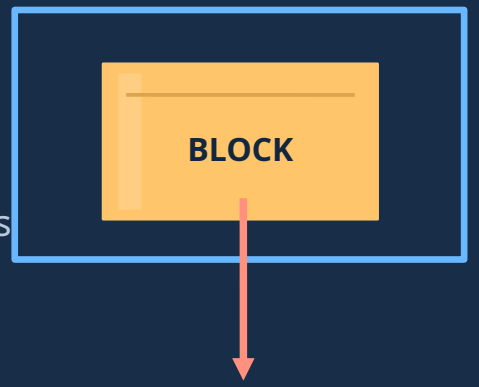
SYSTEMS

First, draw the boundary

An interaction is internal when both participating objects are inside the system.

SYSTEM A • BLOCK ONLY

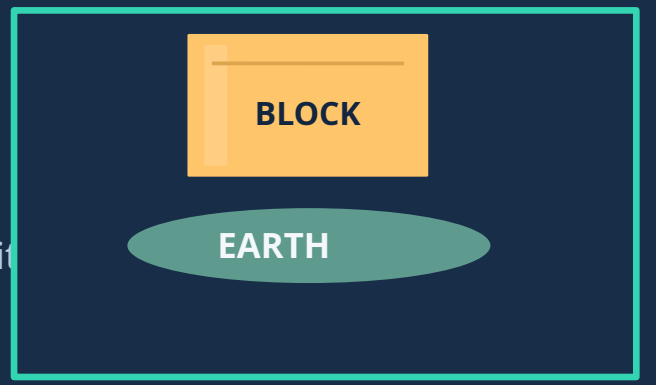
Earth is outs



Gravity: external work

SYSTEM B • BLOCK + EARTH

Their gravit



Use ΔUg instead of Wg

ENERGY STORES

What counts as mechanical energy?

Mechanical energy is kinetic energy + potential energy.

$$E_{mec_h} = K + U$$

K — **Motion**

Kinetic energy

U_g — **Height**

Gravitational potential

U_s — **Extension / compression**

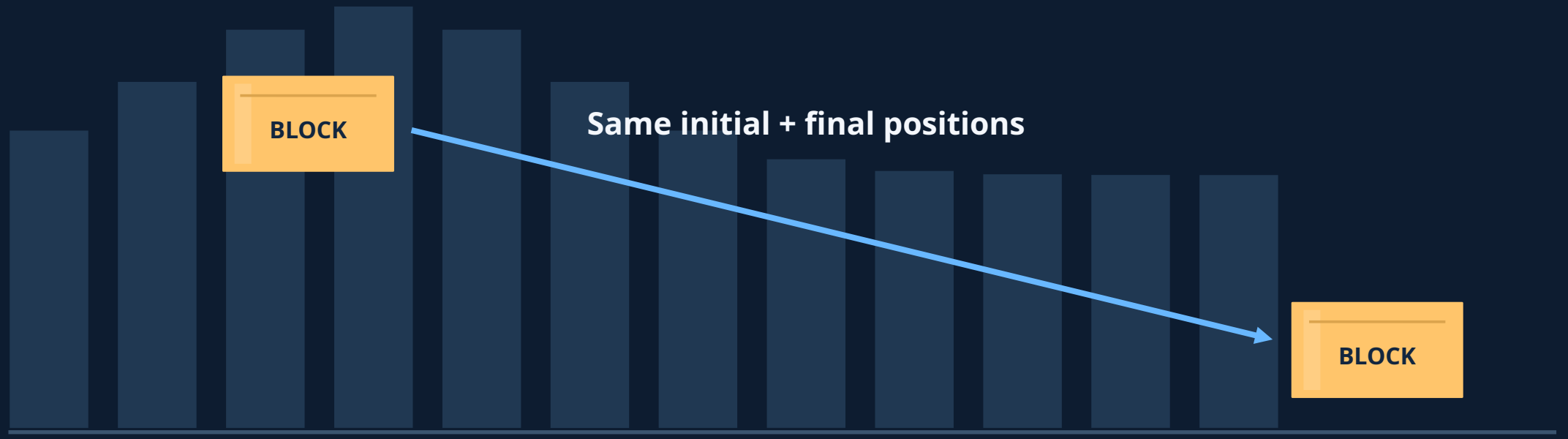
Elastic potential

Internal (thermal) energy is not part of E_{mec_h}.

CONSERVATIVE

A force with a potential-energy store

Path-independent work: $W_{cons} = -\Delta U$



Gravity → Ug = mgh

Ideal spring → $U_s = \frac{1}{2}kx^2$

NONCONSERVATIVE

The route can matter

Sliding friction and drag are nonconservative forces.



For the same f_k, more relative sliding generates more internal energy.

A WORKING METHOD

Three questions before any equation

Define the system first; then select the correct energy terms.

01

WHAT IS THE SYSTEM?

List its physical objects.

02

WHICH ENERGY STORES?

K, Ug and/or U_s?

03

WHICH EXTERNAL WORK?

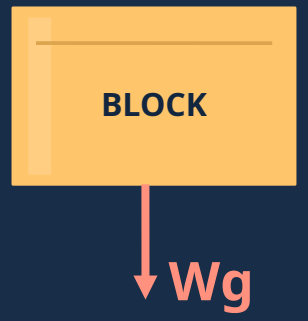
Person, gravity, spring, friction?

AVOID DOUBLE-COUNTING

One interaction. One energy account.

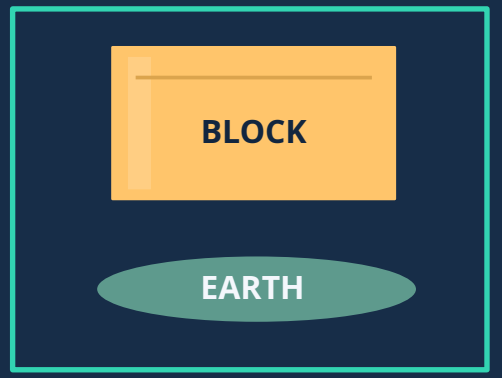
Earth and an ideal spring can be inside or outside — but not both counted twice.

EARTH OUTSIDE



Gravity does external work

EARTH INSIDE



Include ΔUg instead of Wg

No double-dipping!

SYSTEM ENERGY

The full energy account

Use this when you track internal-energy changes within the chosen system.

$$\Delta K + \Delta U + \Delta U_{int} = W_{ext}$$

THIS WORKSHEET

$\Delta K + \Delta U = W_{ext}$

FINER DETAIL

Sliding can generate ΔU_{int}.

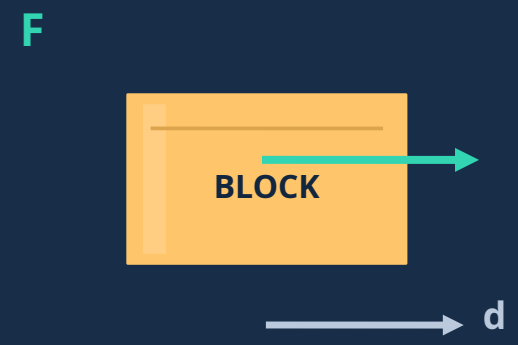
Simplified mechanical-energy calculations do not explicitly track heating within the chosen objects.

SIGN OF WORK

Does the force point with the motion?

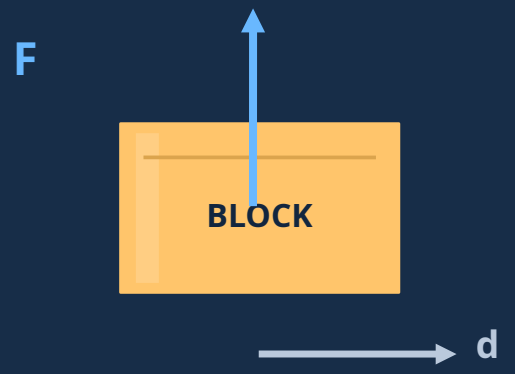
Work = force × displacement × cos(angle)

POSITIVE



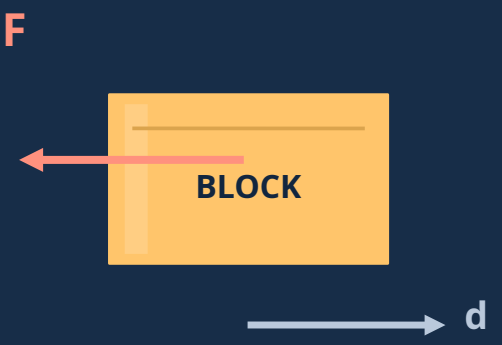
θ = 0° W > 0

ZERO



θ = 90° W = 0

NEGATIVE

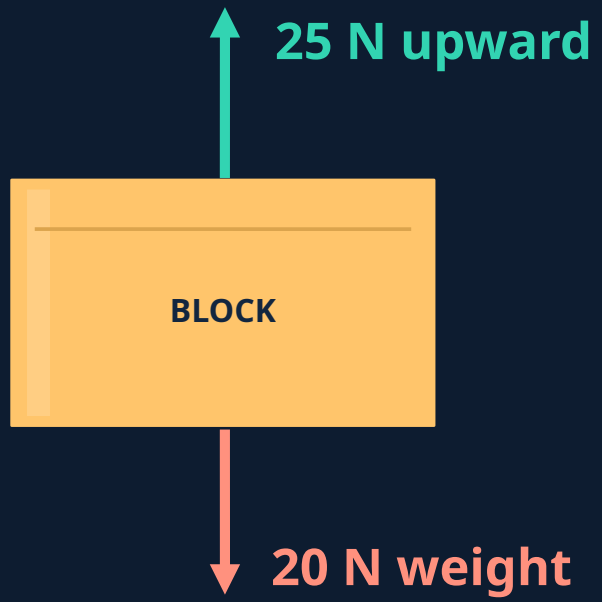


θ = 180° W < 0

WORKSHEET 1

Lifting the block

m = 2.0 kg • d = 0.60 m upward • g = 10 m s^{-2}



PERSON DOES +15 J

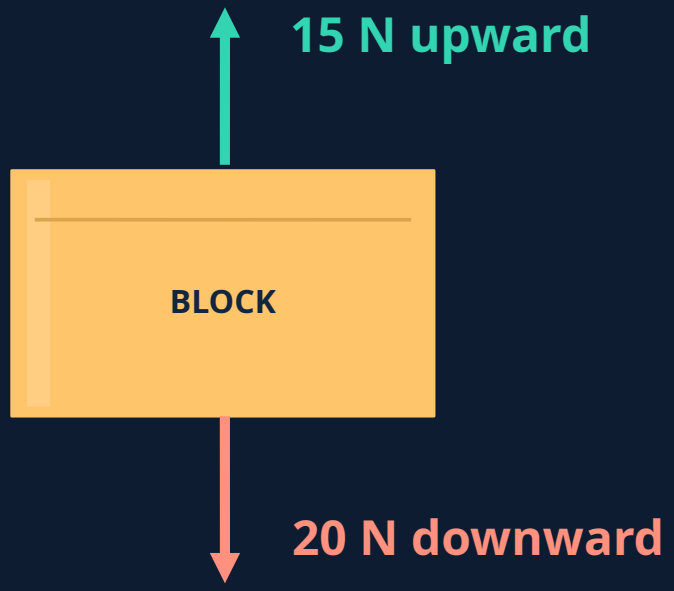
+3 J kinetic

+12 J gravitational

WORKSHEET 2

Lowering the block

The applied force is upward, even though displacement is downward.



GRAVITATIONAL U DECREASES

−12 J

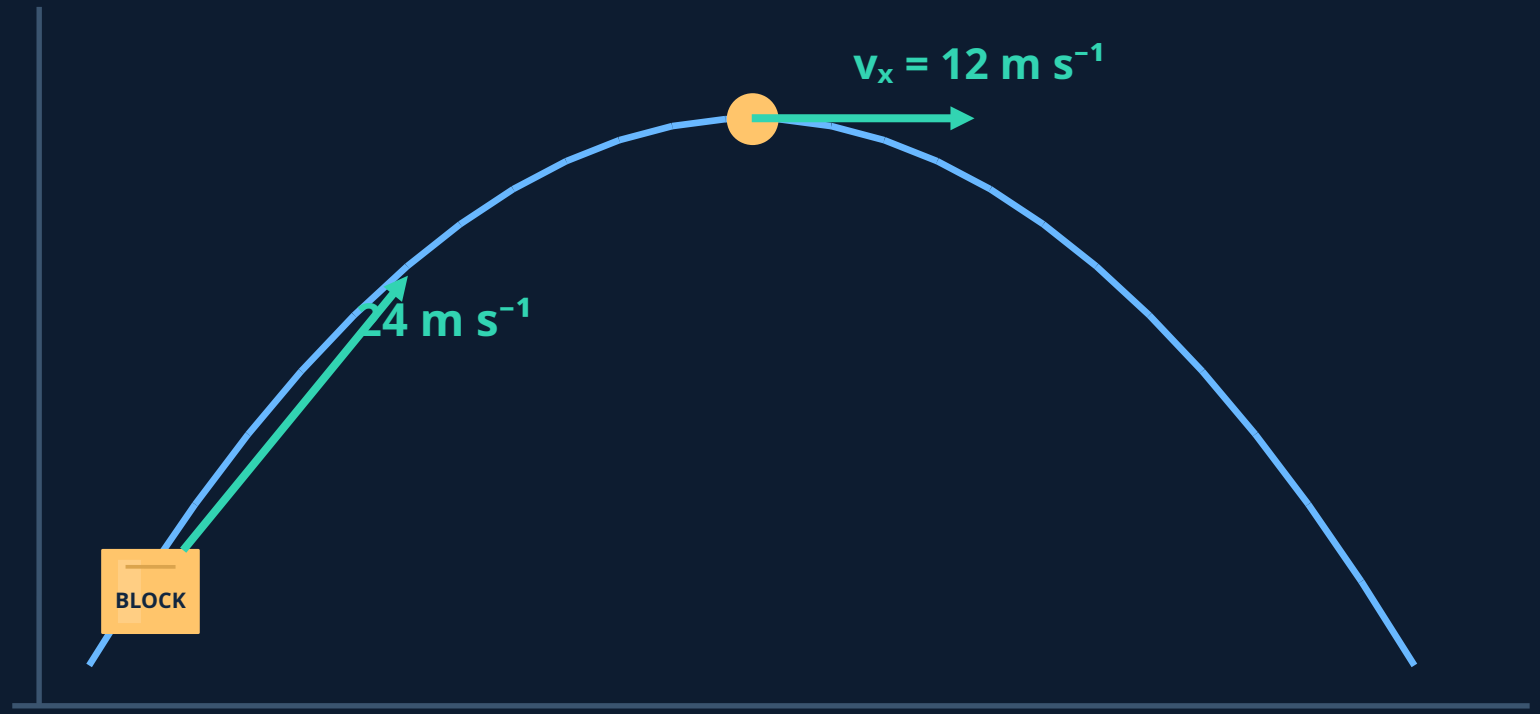
Block gains +3 J of K

Person does −9 J

WORKSHEET 3

At the highest point, speed is NOT zero

Projectile starts at 24 m s^{-1}, 60° above the horizontal; neglect drag.



MAX HEIGHT

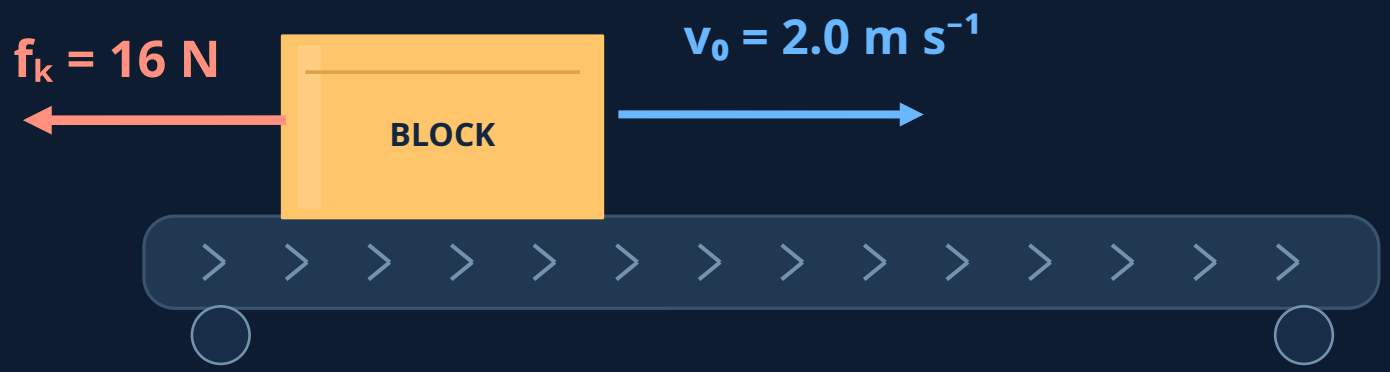
21.6 m

$v_y = 0$ • $v_x \neq 0$

WORKSHEET 4(A)

Kinetic friction brings a block to rest

The belt is already stationary during the sliding interval.



WORK ON BLOCK

-10 J

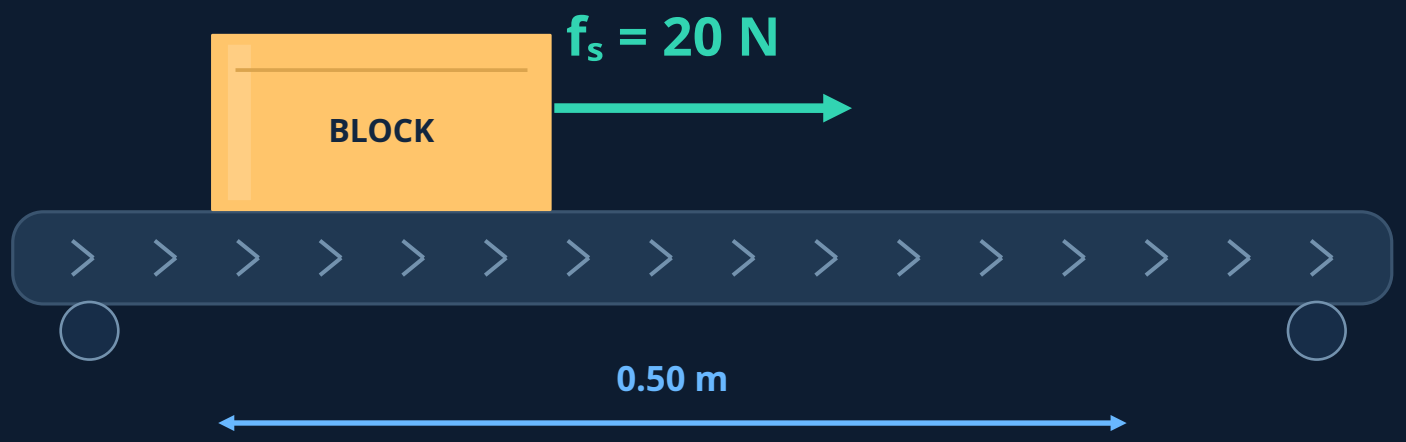
$\Delta U_{int} = +10$ J

Belt displacement = 0

WORKSHEET 4(B)

Static friction can do positive work

Block and belt move together: no relative slipping at their contact.



ON THE BLOCK

+10 J

ON THE BELT: −10 J

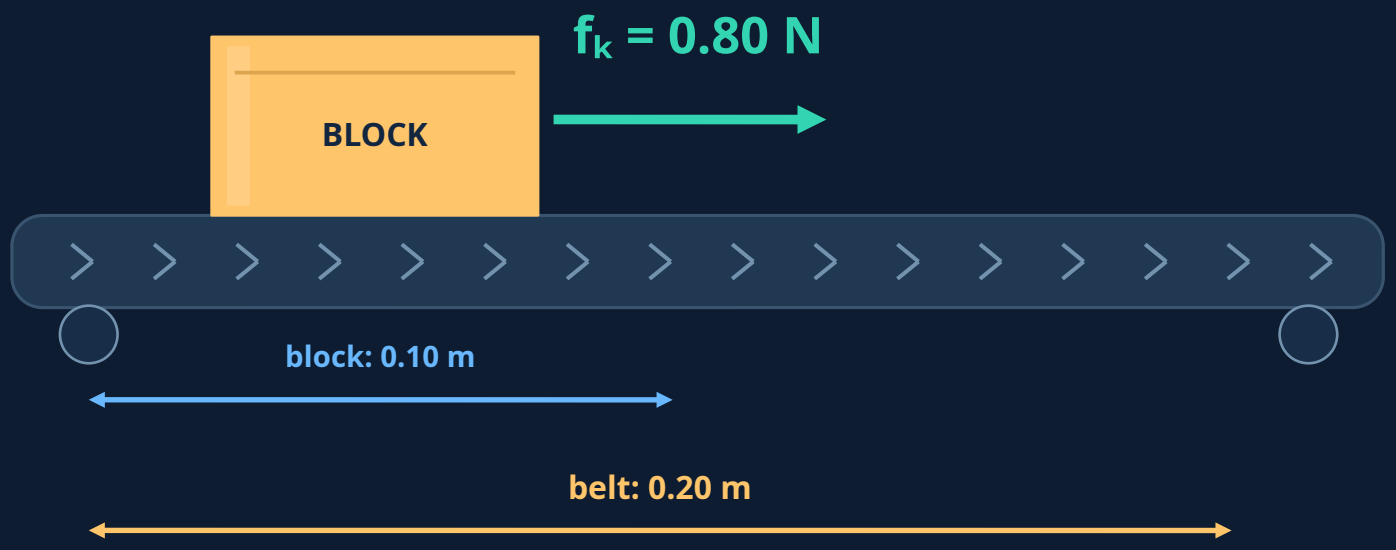
SLIDING DISSIPATION

0 J

WORKSHEET 4(C)

Kinetic friction can also do positive work

Now the belt moves faster than the initially stationary block.



WORK ON BLOCK

+0.080 J

WORK ON BELT

−0.160 J

Block final speed: 0.179 m s^{-1}

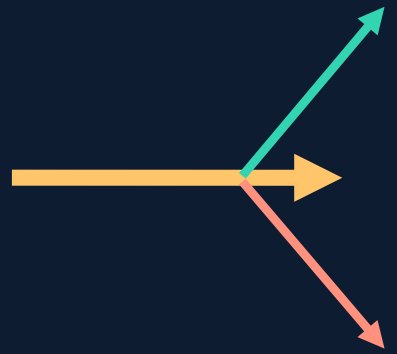
ENERGY FLOW

Where did the belt’s energy go?

The motor replaces the energy that friction removes from the constant-speed belt.

MOTOR / BELT

0.160 J



BLOCK KINETIC ENERGY

+0.080 J

INTERNAL ENERGY AT CONTACT

+0.080 J

$$0.160\ \text{J} = 0.080\ \text{J} + 0.080\ \text{J}$$

TAKEAWAYS

Friction is about relative motion

Its work on one object and its energy dissipation are different quantities.

WORK ON ONE OBJECT

$W_f = f\,d\cos\theta$
May be positive or negative.

SLIDING DISSIPATION

$\Delta U_{int} = f_k d_{rel} \geq 0$
Depends on relative sliding.

EXIT QUESTION → Can kinetic friction do positive work while generating heat?